SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2– CONCEPT MAPPING

Course Code:	CSA6503	**Course Title:**	Generative AI and Large Language Models
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Case study
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)*		
Student Name:	ch.Poorna Teja	**Reg. No.:**	192472149
Section / Batch:	Slot A/2024	**Date:**	04/08/2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __________________________

SECTION A – CONCEPT MAPPING

38. Complete AI Customer Service System

Introduction

Complete AI Customer Service System is an Artificial Intelligence (AI) and Natural Language Processing (NLP) application designed to provide automated customer support by understanding user queries and generating accurate, human-like responses. Traditional customer service often requires human agents to answer repetitive questions, which can be time-consuming, expensive, and difficult to scale. An AI-powered customer service system solves this problem by using Large Language Models (LLMs) to understand customer requests and provide instant assistance.

The system accepts customer queries such as order tracking, product information, return requests, payment issues, account-related questions, and technical support. It processes the user's text, understands the intent, and generates relevant responses based on the given context and instructions. The AI can handle multiple customer conversations simultaneously, reducing waiting time and improving customer satisfaction.

For example, if a customer asks, "My order has not arrived yet. What should I do?", the system can respond by asking for the order number and providing guidance on checking the delivery status. Similarly, if a customer wants to return a product, the AI can explain the return policy and guide the customer through the return process. This makes AI Customer Service Systems highly useful for businesses that receive a large number of customer inquiries every day.

Applications

A Complete AI Customer Service System has many real-world applications:

1. E-Commerce Websites: Provides instant support for order tracking, product information, returns, refunds, and payment-related queries.

2. Banking and Financial Services: Assists customers with account information, transaction details, card services, and general banking inquiries.

3. Healthcare Services: Answers patient questions regarding appointments, hospital services, medical information, and healthcare procedures.

4. Telecommunication Companies: Helps users resolve issues related to mobile plans, internet services, billing, and technical support.

5. Educational Institutions: Assists students by answering questions about admissions, courses, examinations, fees, and academic schedules.

6. Retail and Business Support: Handles customer complaints, product recommendations, warranty information, and service requests efficiently.

Examples

Example 1

Input: "Where is my order? It has not been delivered yet."

Output:
"I'm sorry for the inconvenience. Please provide your order number so I can check the delivery status and assist you further."

Example 2

Input: "I want to return my headphones."

Output:
"Sure! Please share your order number and the reason for the return. I'll guide you through the return process."

Example 3

Input: "How can I reset my account password?"

Output:
"You can reset your password by clicking the 'Forgot Password' option on the login page and following the instructions sent to your registered email address."

Example 4

Input: "Do you have laptops under ₹50,000?"

Output:
"Yes, we have several laptops under ₹50,000. Please let me know your preferred brand or specifications, and I'll recommend suitable options."

Thus, a Complete AI Customer Service System provides a fast, intelligent, and cost-effective solution for handling customer inquiries. It reduces manual workload, improves response time, enhances customer satisfaction, and enables businesses to provide 24/7 automated support across various industries.

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The system takes customer queries as input and processes the textual information using a Large Language Model (LLM). The model understands the meaning, context, and intent of the query and generates an appropriate response. It can answer frequently asked questions, provide troubleshooting guidance, assist with product information, and support customers throughout the conversation.

For example, if a customer asks, "My order has not arrived yet. What should I do?", the system can respond by asking for the order number and providing guidance on checking the delivery status. Similarly, if a customer wants to return a product, the AI can explain the return policy and guide the customer through the return process. This makes AI Customer Service Systems highly useful for businesses that receive a large number of customer inquiries every day.

Applications

Complete AI Customer Service System has many real-world applications:

1. E-Commerce Websites: Provides instant support for order tracking, product information, returns, refunds, and payment-related queries.

2. Online Shopping Platforms: Helps customers find products, compare items, and receive personalized recommendations.

3. Banking and Financial Services: Assists customers with account information, transaction details, card services, and general banking inquiries.

4. Healthcare Services: Answers patient questions regarding appointments, hospital services, medical information, and healthcare procedures.

5. Educational Institutions: Assists students by answering questions about admissions, courses, examinations, fees, and academic schedules.

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Thus, Complete AI Customer Service System provides a fast, intelligent, and cost-effective solution for handling customer inquiries using Artificial Intelligence and Natural Language Processing. It reduces manual effort, improves response time, enhances customer satisfaction, and enables businesses to provide 24/7 automated customer support across various industries.

2 PEFT Strategy Selection

Introduction

PEFT (Parameter-Efficient Fine-Tuning) Strategy Selection is a technique in Artificial Intelligence (AI) and Natural Language Processing (NLP) used to efficiently adapt Large Language Models (LLMs) to specific tasks by updating only a small subset of model parameters instead of retraining the entire model. Traditional fine-tuning requires modifying millions or even billions of parameters, which demands significant computational resources, memory, and time. PEFT overcomes this challenge by using lightweight adaptation strategies such as **LoRA (Low-Rank Adaptation), Prefix Tuning, Prompt Tuning, P-Tuning, and Adapter Layers**.

The system takes task-specific information such as the dataset, task type, model size, available hardware, memory limitations, and performance requirements as input. Based on these factors, it selects the most suitable PEFT strategy that provides the best balance between accuracy, training speed, memory usage, and computational cost.

For example, if a user wants to fine-tune a Large Language Model for sentiment analysis on a GPU with limited memory, the system may recommend **LoRA** because it offers high accuracy while requiring only a small number of trainable parameters. Similarly, if the task involves prompt-based text generation, the system may recommend **Prompt Tuning** or **Prefix Tuning**. This makes PEFT Strategy Selection highly useful for efficiently adapting foundation models in real-world applications.

Applications

PEFT Strategy Selection has many real-world applications:

1. **Large Language Model Fine-Tuning:** Selects efficient fine-tuning methods for LLMs such as Llama, GPT, and Gemma.
2. **Natural Language Processing:** Optimizes models for tasks such as text classification, summarization, translation, and question answering.
3. **Computer Vision:** Fine-tunes vision transformer models for image classification and object detection with minimal computational cost.
4. **Healthcare AI:** Adapts medical AI models using limited healthcare datasets while reducing hardware requirements.
5. **Chatbots and Virtual Assistants:** Efficiently customizes conversational AI models for specific business domains.
6. **Research and Education:** Helps researchers and students fine-tune pre-trained models using affordable computing resources.

Examples

Example 1:

Input: Fine-tune Llama 3 for sentiment analysis using a single GPU with limited memory.

Output: Recommended Strategy → **LoRA (Low-Rank Adaptation)**

Example 2:

Input: Customize a language model for text generation with minimal trainable parameters.

Output: Recommended Strategy → **Prompt Tuning**

Example 3:

Input: Fine-tune a BERT model for question answering using limited computational resources.

Output: Recommended Strategy → **Adapter Layers**

Example 4:

Input: Adapt a language model for summarization while preserving the original model weights.

Output: Recommended Strategy → **Prefix Tuning**

Thus, **PEFT Strategy Selection** provides a fast, efficient, and cost-effective approach for fine-tuning Large Language Models using minimal computational resources. It reduces training time, lowers memory consumption, preserves the original model parameters, and enables organizations to deploy customized AI models efficiently across various real-world applicatio**n**

Examples

Example 1:
Input: *"Write a C++ program to check whether a number is even or odd."*
Output: The assistant generates a C++ program using conditional statements.

Example 2:
Input: *"Create a Python program to calculate the factorial of a number."*
Output: The assistant generates Python code using a loop or function.

Example 3:
Input: *"Find the error in this program and correct it."*
Output: The assistant identifies the programming error and provides the corrected code.

Conclusion

A Code Generation Assistant is a useful AI tool that reduces programming effort and improves developer productivity. It can generate, explain, debug, and modify code based on user requirements. It is particularly useful for students, beginners, and software developers. However, generated code should always be reviewed and tested because AI-generated code may sometimes contain logical or syntax errors.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical / Concept	30%	All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships	Most concepts correct; minor gaps in relationships	Several concepts missing or misclassified	Major conceptual errors; map incomplete
Application	25%	Real-world examples linked accurately to each concept	Examples mostly relevant	Few or generic examples	No real-world linkage shown
Quality	20%	Logical, well-organized structure with clear connectors	Mostly organized; minor structural issues	Some disorganization affecting clarity	Poorly structured, hard to follow
Presentation	15%	Visually clear, creative, easy to interpret	Neat and readable	Cluttered or inconsistent formatting	Untidy, difficult to interpret
Documentation / Communication	10%	Concise labels/notes that add clarity	Adequate labeling	Minimal or unclear labeling	No supporting labels or explanation

Marks Evaluation:

Question No.	Technical / Concept(30%)	Application(25%)	Quality (20%)	Presentation (15%)	Documentation / Communication (10%)	Total Marks (Each – 50)
Q1						
Q2						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________